



# TECHNEX'26

13 MAR- 15 MAR 2026

## MLWARE Sherlock Files



# OVERVIEW

You are Sherlock Holmes, the world's most perceptive investigator of patterns, motion, and hidden cause-effect relationships. A peculiar case has just arrived at 221B Baker Street. A courier delivers a sealed envelope marked "Evidence – Motion Records." Inside, you find a collection of video fragments recovered from a recent crime scene. The recordings appear ordinary at first glance—objects falling, colliding, sliding, bouncing, and interacting according to the laws of motion. But something is wrong. The files have clearly been tampered with. The frames that once recorded each event in a precise sequence have been scrambled, leaving only scattered moments frozen in time. What should have been a clear chain of cause and effect has become a puzzle of disconnected clues. Each video now exists as a set of shuffled frames. Somewhere within this disorder lies the true timeline of events. Your task is to build a deep learning model that reconstructs the correct chronological order of these shuffled frames and restores the true sequence of motion. By carefully analyzing the evidence, you must reconstruct the correct chronological order of the frames and restore the true sequence of motion. In Sherlock's Files, every frame is a clue. Only by piecing them together can you uncover what really happened and decode the dynamics hidden within the motion.

# STRUCTURE

The competition has a total of 2 rounds where the 1st will be conducted online while the 2nd round will be conducted offline.

## Round 1 : MODEL SUBMISSION

Objective: Participants must develop a fully functional deep learning model that can reorder the frames of videos in the given dataset.



## DATASET STRUCTURE

The dataset consists of short video clips extracted from action videos. Each clip contains at most 150 frames and represents a continuous motion sequence.

- To create the challenge, the original videos have been intentionally corrupted by modifying their temporal structure. Participants are required to recover the correct chronological order of frames.

The dataset is divided into two subsets:

dataset/

| ->train/

| ->video\_0.mp4

| ->video\_1.mp4

| ->...

| ->test/

| ->video\_5000.mp4

| ->video\_5001.mp4

| ->...

| ->train\_labels.json

## LABEL FORMAT

The labels are provided in JSON format.

Example:{

"video\_0": [5, 6, 7, 8, 9, 4, 3, 2, 1, 0]

}

- This list specifies the order in which frames from the corrupted video must be selected to reconstruct the correct chronological sequence.
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Each number represents the position of a frame in the corrupted video.

The reconstruction works as follows:

1st frame → take the frame at position 5

2nd frame → take the frame at position 6

3rd frame → take the frame at position 7

4th frame → take the frame at position 8

.....

Thus, the JSON label specifies how the frames of the corrupted video must be reordered to recover the true temporal sequence.



## DATASET SIZE

The dataset contains approximately: 5000 training videos & 100 test videos  
Each video contains between 5 and 150 frames depending on the original clip length.

## SUBMISSION FORMAT

Participants must submit their predictions in a CSV file with the following structure:

video\_id,order

video\_5000,"5 6 7 8 9 4 3 2 1 0"

video\_5001,"3 4 5 0 1 2"

Each row corresponds to one test video.

- video\_id – the identifier of the test video
- order – a space-separated list representing the predicted frame ordering

The numbers represent the positions of frames in the corrupted video.

The sequence describes the order in which frames should be selected to reconstruct the correct chronological sequence.

## EVALUATION METRIC: KENDALL RANK CORRELATION (KENDALL'S TAU)

### Example

Ground truth:

`{"video_0": [2, 3, 4, 1, 0]}`

Prediction:

`{"video_0": [2, 4, 3, 1, 0]}`

All frame pairs (10 total):

(2,3) (2,4) (2,1) (2,0) (3,4) (3,1) (3,0) (4,1) (4,0) (1,0)

### Comparison:

Correct pairs = 9 , Incorrect pairs = 1

Kendall Tau formula:

$$\tau = (C - D) / (C + D)$$

where, C = number of concordant pairs, D = number of discordant pairs

Calculation:

$$\tau = (9 - 1) / (9 + 1)$$

$$\tau = 8 / 10$$

$$\tau = 0.8$$

Final leaderboard score = average Kendall Tau across all test videos.



## Final Submission:

All files must be uploaded in a single ZIP folder containing:

1. Model Weights
2. Working Codebase
3. Inference Script
4. A one-page document explaining the rationale behind the approach
5. requirements.txt or environment.yml for reproducibility

## Naming Format:

TeamName\_ModelRound

## Submission Process:

- Submissions will be accepted through the official Google Form.
- A separate Kaggle competition will also be hosted where participants must submit predictions for the test set.
- The Kaggle leaderboard will be used to track performance and rankings.

## Important Rules

- API calls are not allowed.
- External datasets are allowed.
- Pretrained open-source models are allowed.
- Although there is no strict limit on model size, participants are encouraged to design efficient and lightweight models with fast inference pipelines.
- We encourage participants to develop physics-aware and interpretable deep learning approaches that capture the underlying dynamics of motion in the videos. Solutions that rely purely on simple heuristics or frame similarity may not generalize well. Instead, methods that incorporate principles of temporal reasoning, physical consistency, and interpretable modeling are strongly encouraged.



# Round 2 : MODEL PRESENTATION

Teams that make a submission in Round 1 will be invited to present their solutions before the judging panel.

Qualified teams must prepare and submit a presentation deck (PDF or PPT) describing their approach.

The presentation should cover the following components:

1. Problem Understanding
2. Model Architecture and Methodology
3. Feature Engineering and Data Processing
4. Evaluation Strategy
5. Results and Insights

Presentation Format:

Each team will be given:

- 10 minutes for presentation
- 5 minutes for questions from the judges

Presentations will be conducted in offline mode

Marking Scheme :

Evaluation will be based on the following criteria:

1. Innovation - 20%
2. Accuracy (Kendall Score) - 30%
3. Model Efficiency / Size - 20%
4. Code & Report Clarity - 10%
5. Presentation - 20 %

Kaggle competition link ( Includes the dataset ) -

<https://www.kaggle.com/competitions/ml-ware-26-sherlock-files>

Whatsapp Group Link -

[https://chat.whatsapp.com/LdYunj3VbUx3zobedB1NxJ?mode=gi\\_t](https://chat.whatsapp.com/LdYunj3VbUx3zobedB1NxJ?mode=gi_t) Join the participants group